

STUDY PLAN

Juvenile Chinook Salmon and Steelhead Survival through the Priest Rapids Project in 2026

PREPARED BY:

Public Utility District No. 2 of Grant County

LGL Environmental Services Inc.

PREPARED FOR:

Priest Rapids Coordinating Committee

Version History:

Date:

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1.0 Introduction

The Priest Rapids Project (hereafter the Project) is owned and operated by the Public Utility District No. 2 of Grant County (Grant PUD). The Project includes two hydroelectric developments, Wanapum and Priest Rapids dams.

On April 17, 2008, the Federal Energy Regulatory Commission (FERC) issued its Order Issuing New License for the Priest Rapids Project to Grant PUD (FERC No. 2114). Incorporated into the New License was Term and Condition 1.1 – NMFS 2008 Biological Opinion for the Priest Rapids Project (NMFS 2008) (adapted from Action 1, NMFS 2004) and 401 Water Quality Certification Section 6.4.6.a.

FERC shall require Grant PUD to make steady progress toward achieving a minimum 91% combined adult and juvenile salmonid survival performance standard at the Priest Rapids and Wanapum developments (i.e., each dam). The 91% standard includes a 93% project-level (reservoir and dam) juvenile performance standard. NMFS recognizes that it is not currently possible to measure the 91% combined adult and juvenile survival standard. Grant PUD shall therefore continue to conduct dam and reservoir smolt survival studies, evaluating progress toward meeting 93% juvenile Project passage survival. This standard can be measured at each development individually, or as a composite of survival at the two developments.

NMFS recognizes that the juvenile standard has already been achieved for UCR spring-run Chinook salmon. FERC shall require Grant PUD to at least maintain this level of survival. FERC shall ensure that Grant PUD achieves the juvenile standard for UCR steelhead, as measured after 3 consecutive years of evaluation, by 2013. Grant PUD can compensate for a failure to achieve the performance standard at one of its developments by exceeding the performance standard at the other development (i.e., at a minimum, by the same percentage amount below the survival performance standard at the development failing to meet performance standards). If Project survival exceeds the minimum combined juvenile and adult performance standard specified above, as measured per the specifications listed below, off-site mitigation obligations can be reduced by a commensurate amount.

In 2003-2005, Grant PUD released radio- and PIT-tagged juvenile anadromous salmonids (e.g., yearling Chinook Salmon) as part of the continuing effort to monitor and evaluate survival of fish migrating through the Project (Anglea et al. 2004, 2005a,b). Acoustic tag studies were first conducted in 2004 and were employed in additional studies throughout the Project in 2006-2011 and 2014-2017 (Timko et al. 2007, 2008, 2010, 2011; Sullivan et al. 2009; Thompson et al 2012; Hatch et al. 2015, 2016, 2017, 2018).

Performance standards for juvenile steelhead were evaluated for three consecutive years (2008-2010), and, as outlined in the NMFS 2008 Biological Opinion (BiOp), must average 86.49% or higher to meet the licensed agreements. The performance standards for steelhead in 2008-2010 was 81.05%, which fell short of the performance standard goals. Three more consecutive steelhead survival evaluation studies were conducted (2015-2017), with the arithmetic 3-year average of 87.0% (Hatch et al. 2016, 2017, 2018). This 3-year survival estimate was presented to the PRCC for approval and accepted in December 2017 (PRCC-SOA 2017-07). The next scheduled steelhead “check-in” survival study was scheduled for 2022 (PRCC-SOA 2017-02, item #8). In January of 2018, PRCC-SOA 2018-01 was presented to the PRCC, modifying the juvenile survival “check-in” survival studies for yearling Chinook Salmon, Sockeye Salmon, and steelhead with the first check-in study slated to occur in 2025, with subsequent “check-in” studies happening every 10 years thereafter. This SOA (PRCC_SOA 2018-01) was approved January 31, 2018.

The 2025 “check-in” survival study was subsequently delayed by one year due to the inability to secure a helicopter contractor, a critical component of the established study design. As a result, the study will now proceed in the spring of 2026 to evaluate the survival of juvenile steelhead and yearling Chinook Salmon through the Priest Rapids Project. Following this effort, a survival evaluation for juvenile Sockeye Salmon is scheduled for 2027, with 2028 identified as

a contingent evaluation year, if necessary, to address any data gaps or anomalies arising from the 2026 or 2027 studies. Grant PUD has contracted LGL Environmental Services Inc. (LGL) to conduct the 2026–2028 survival evaluations and Buchanan Statistical Consulting LLC to conduct the statistical analyses.

2.0 Objectives

This document describes the proposed plan to conduct a survival evaluation through the Priest Rapids Project Area during the 2026 spring smolt out-migration using acoustic-telemetry techniques.

The following tasks will be addressed by the 2026 survival evaluation:

Task 1) Estimate juvenile steelhead and yearling Chinook Salmon survival and the associated error through the Priest Rapids Project using a paired-release model during the 2026 spring outmigration;

Task 2) Estimate migration rates, reach mortality, and arrival distributions for juvenile steelhead and yearling Chinook Salmon through the Wanapum and Priest Rapids developments during the 2026 spring outmigration;

Task 3) Support the evaluation of avian predation impacts on juvenile steelhead within the Priest Rapids Project by providing tagging and telemetry data to RTR/OSU to aid in their analysis of PIT tags recovered at avian nesting colonies on the Mid-Columbia Plateau.

3.0 Site Description

3.1 *Priest Rapids Project*

Public Utility District No. 2 of Grant County, Washington (Grant PUD) owns and operates two hydroelectric dams on the Columbia River—Wanapum and Priest Rapids—known altogether as the Priest Rapids Hydroelectric Project (PRP) and is operated under the terms and conditions of the Federal Energy Regulatory Commission (FERC) Hydroelectric License No. P-2114 issued by FERC on April 17, 2008.

Grant PUD operates the PRP through the coordinated operation of the seven-dam system and other Columbia Basin entities with current operational agreements with the fishery agencies and other operators to provide protection and improvement for a range of fisheries and other resources within and downstream of the PRP. These agreements include the Hanford Reach Fall Chinook Protection Program Agreement (HRFCPPA, Grant PUD 2004), the Priest Rapids Project Salmon and Steelhead Settlement Agreement (SSSA, Grant PUD 2006). The PRP is also subject to the requirements of its FERC License and related laws and regulations, as well as to the requirements (incorporated by reference in the license) of the Biological Opinion (BiOp) of the PRP issued by the National Marine Fisheries Service (NMFS) for its effects on anadromous salmonids, the Clean Water Act Section 401 Water Quality Certification (WQC) issued by the Washington Department of Ecology (WDOE), and the BiOp for the PRP issued by the United States Fish and Wildlife Service (USFWS) regarding the effect of the PRP on Bull Trout.

3.2 *Wanapum Dam*

Wanapum Dam is located on the Columbia River at river mile (RM) 416, 19 miles upstream of Priest Rapids Dam, and 38 miles downstream from Rock Island Dam. The dam has a 1,540 ft long powerhouse, oriented approximately perpendicular with river flow, and an 832 ft long spillway at approximately a 45° angle to flow. The powerhouse has 10 Kaplan turbine units, numbered from north to south, with a combined generating capacity of 1,038 megawatts. There are additional intake structures for six potential future units. Each turbine unit has three intake slots, designated A, B and C from north to south.

The Wanapum Dam spillway contains 12 tainter gates, each 65-ft tall by 50-ft wide. These gates open at the bottom to produce submerged spill at a depth of 65 ft. The Wanapum Fish Bypass (WFB) is 20 ft wide and normally spills 20 kcfs of surface water. The WFB is located at Future Unit 11. At a forebay elevation of 570 ft the WFB can operate to pass an estimated 2, 5, 10, 15, or 20 kcfs. (Oakwood Consulting Inc., 2005)

3.3 Priest Rapids Dam

Priest Rapids Dam is located at RM 397 and is 105 miles upstream of McNary Dam. The dam is 19 miles downstream of Wanapum Dam and approximately 70 miles upstream from the confluence of the Snake and Columbia rivers. Normal maximum forebay elevation is 488 ft above MSL and the normal operating head is 78 ft.

The dam has a 1,025 ft long powerhouse located at the northeast end, oriented approximately perpendicular to river flow, and a 1,152 ft long spillway on the southwest end of the dam. The powerhouse has 10 Kaplan turbine units, numbered from west to east, with a combined nameplate generating capacity of 956 megawatts. Each turbine unit has three intake slots, designated A, B, and C from east to west.

The spillway contains 22 tainter gates, each 50 ft tall by 40 ft wide. These gates open at the bottom to produce submerged spill at a depth of 50 ft. Each spill bay passes an average of 8-10 kcfs under normal operating conditions. The Priest Rapids Fish Bypass (PRFB) was constructed at spillbays 20, 21, and 22 and became operational in 2014 as a bypass structure to downstream migrating smolts; the bypass operates as three independent surface-spill gates, each passing up to 9 kcfs.

4.0 Study Design

The methods used to evaluate juvenile steelhead and yearling Chinook Salmon survival through the Priest Rapids Project in 2026 will be similar to those employed during the 2015-2017 Grant PUD survival evaluations (e.g., Hatch et al. 2017). The study will be developed and conducted to achieve a standard error (SE) no greater than $\pm 2.5\%$. The identified standard error ($SE \pm 2.5\%$) and model assumptions originally developed by J. Skalski (University of Washington; School of Aquatic and Fishery Science) (see Appendix A) will be used, as required by PRCC-SOA 2017-02, item #6.

Task 1: Estimate Project Survival

The Priest Rapids Project passage survival will be estimated using a paired-release model, following the methods of Skalski et al. (2014). The general paired-release methodology of this 2026 study will estimate Project Survival using a similar approach to the survival studies conducted at both developments in 2008-2010, and 2014-2017 (Skalski et al. 2009, 2010a, b; 2014, 2016, 2017, 2018).

Paired releases of 1,800 run-of-river yearling Chinook Salmon and 1,800 juvenile steelhead (Table 1) will be collected and implanted with acoustic transmitters (ATS SS-400 tag) and released in the tailrace of Rock Island Dam (n=1,000; treatment group) and Priest Rapids Dam (n=800; control group).

To test the key assumption of the survival model that terminal arrays are positioned sufficiently downstream to prevent the detection of drifting mortalities, fifty dead fish from each species will be implanted with acoustic tags and released into the Priest Rapids Dam tailrace alongside the release of live tagged study fish.

Table 1. Estimated quantity juvenile steelhead and yearling Chinook Salmon to be tagged and released during the 2026 Priest Rapids Project survival evaluations. *All dead fish will be released into the Priest Rapids Dam Tailrace. **All life test tags will be randomly selected and tested in a single pool of 100 to produce a life test curve that represents the acoustic tags deployed in 2026 (Appendix A).

Release Location	Yearling Chinook	Juvenile Steelhead
<i>Rock Island Dam Tailrace</i>	1,000	1,000
<i>Priest Rapids Dam Tailrace</i>	800	800
<i>Dead Fish Detection Tests*</i>	50	50
<i>Tag Life Testing**</i>	50	50
Total Tags	1,900	1,900

As displayed in Figure 1, the treatment group of study fish (released in the tailrace of Rock Island Dam) will be monitored as they migrate through the Project area at the boat restricted zone hazard barriers (hereafter BRZ) of both dams at Wanapum Dam (RM 416) and Priest Rapids Dam (RM 397), along with detection arrays located at Crescent Bar (RM 440), Sunland Estates (RM 430), and Mattawa (RM 408).

The terminal detection arrays will be utilized to detect both treatment (Wanapum Dam) and control (Priest Rapids Dam) released study fish to estimate Project survival. These arrays will be deployed at Lower Ringold (RM 351) and Hanford (RM 339) which are 46 and 58 river miles, respectively, downstream of Priest Rapids Dam. This distance is intended to provide sufficient time for any handling effects to manifest within the control release group, thereby ensuring the assumptions of the survival model are met. In previous steelhead survival studies (Skalski et al. 2016), it was observed that placing the terminal detection arrays too close to the control release site did not allow adequate time for tagging and handling effects to manifest within the control group, potentially violating key model assumptions.

Also, the distance traveled by treatment fish passing through Priest Rapids Dam to the terminal arrays is intentionally designed to prevent the detection of passage-related mortalities. This assumption will be tested through programmatic dead fish releases below Priest Rapids Dam, which are designed to verify that the terminal arrays are positioned far enough downstream to exclude detections of non-surviving fish.

To verify the operating life of the acoustic tags and generate tag-life correction curves for the survival model, a battery-life test will be conducted using 100 acoustic tags randomly selected from the pool of available tags. These tags will be placed in a holding tank with continuously cycled river water, maintained at ambient river temperatures to simulate natural environmental conditions, consistent with methods used during the 2014–2017 studies (Hatch et al. 2015, 2016, 2017, 2018).

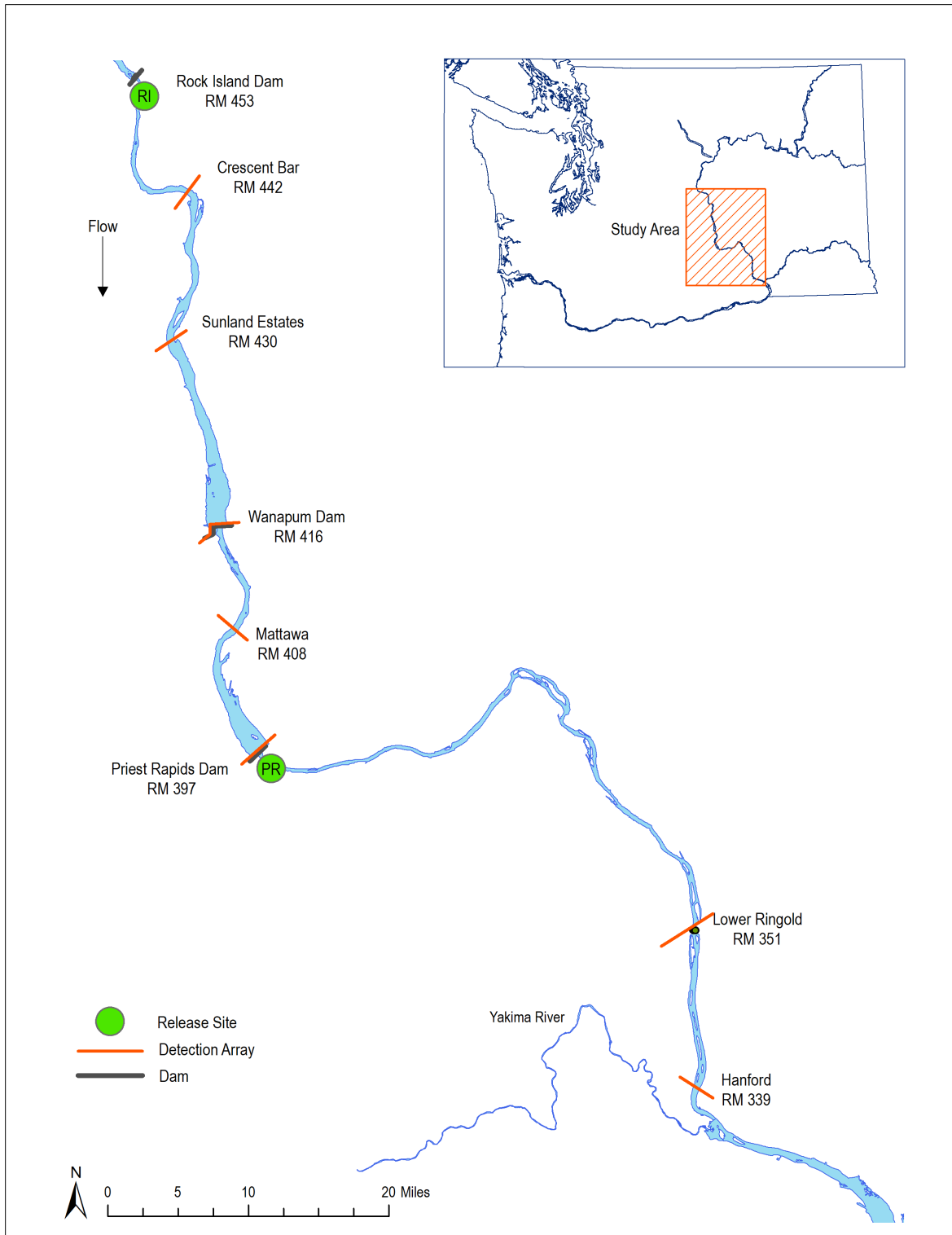


Figure 1. A map depicting the locations of Wanapum and Priest Rapids dams, along with the acoustic array and release locations in the mid-Columbia River, Washington for the 2026 juvenile steelhead and yearling Chinook Salmon survival evaluations.

Task 2: Estimate Migration Rates, Reach Mortality, and Arrival Distributions of Downstream Migrants

The migration rates, reach mortality rates, and BRZ arrival distributions of downstream migrating juvenile steelhead and yearly Chinook Salmon will be estimated through both hydropower developments (Wanapum and Priest Rapids) similar to Hatch et al. (2015, 2016, 2017, 2018).

Task 3: Support Avian Predation Research by RTR

All study fish will be double-tagged with an acoustic tag (ATS SS-400) and a passive integrated transponder (PIT) tag. Post-nesting season, Caspian Tern colonies at Goose Island (Potholes) and Crescent Island (McNary forebay) and other colonies determined by RTR will be scanned for PIT tags deposited during the 2026 nesting season. Any detected PIT tags will be compared to the master list of PIT tags released during this study. Additionally, by using acoustic tag detection histories, a “last detect” will be used to determine the approximate section of river within the Project where the predation event likely occurred.

4.1 Acoustic Transmitters

Consistent with the 2015-2017 studies, all study fish will be double tagged with an acoustic tag (ATS Model SS-400 acoustic tag) and a PIT tag (Biomark 12 mm PIT). Tags will be surgically implanted into run-of-river juvenile steelhead and yearling Chinook Salmon. Upon delivery, all acoustic tags will be randomly distributed into a release group (i.e., treatment, control, dead fish, or life test) and paired with a PIT tag.

4.2 Acoustic Detection System

Autonomous acoustic receivers from two vendors (*Advanced Telemetry Systems [ATS]* and *Teknologic*) will be utilized to monitor and record the presence of acoustic tagged fish during the 2026 study. Acoustic receivers will be deployed using one of three methods, either a near-shore ‘cage mount setup’, a ‘BRZ setup’ when hung to the hazard barriers at Wanapum or Priest Rapids dams, or an acoustic release ready ‘offshore setup’ (Figure 2).

During cage-mounted deployments, a receiver is affixed within a weighted steel cage structure, and then tethered to shore by wire rope (Figure 2). After tethering to a shore-based fixed object, the cage-mounted assembly is and then dropped from the boat into the river about 200 m off shore. The cage mount acts as an anchor that affixes the receiver to the river bottom, while also providing protection to the receiver from debris. Cage-mounted receivers are retrieved by recovering the wire rope from the tethered location, securing it within a heavy-duty pulley attached to the boat, and then backing away from shore using the boat to pull up the cage mount and receiver. The benefit to this deployment method is that the receiver is both accessible to researchers and inaccessible to the public. When possible, receivers nearest the left and right banks at Crescent Bar, Sunland Estates, Mattawa, Lower Ringold, and Hanford¹ will be deployed using shore-based cage mounts (Figure 3 & Figure 4).

The second deployment method is the ‘BRZ setup’ where a weighted receiver is hung from a buoy or cable on the Boat Restricted Zone (BRZ or hazard barrier) at Wanapum and Priest Rapids dams (Figure 2). This deployment method allows receiver accessibility but is only possible on the BRZ buoy lines at Wanapum and Priest Rapids dams (Figure 5).

The final deployment method is the ‘offshore setup’; where the receiver is attached to a *Teledyne* acoustic release and anchored to the river bottom (Figure 2). This setup allows for the deployment of receivers on the mid-river portions (i.e., further than 200 m from shore) to complete the arrays at Crescent Bar, Sunland Estates, Mattawa, Lower Ringold, and Hanford (Figure 3 & Figure 4). Receivers are eventually recovered by activating the coded acoustic release device,

¹ A shore-based cage mount deployment on Hanford right bank will not be possible due to the inaccessibility of the Hanford site.

which detaches the receiver from the anchor, allowing it to float to the surface for recovery. This setup is inaccessible to the public but is not considered accessible mid-study, because the release device needs to be activated to retrieve the receiver, and the setup would need to be re-anchored should re-deployment occur. Communication with these receivers is possible during the study to ensure that they are present and operating within expectations.

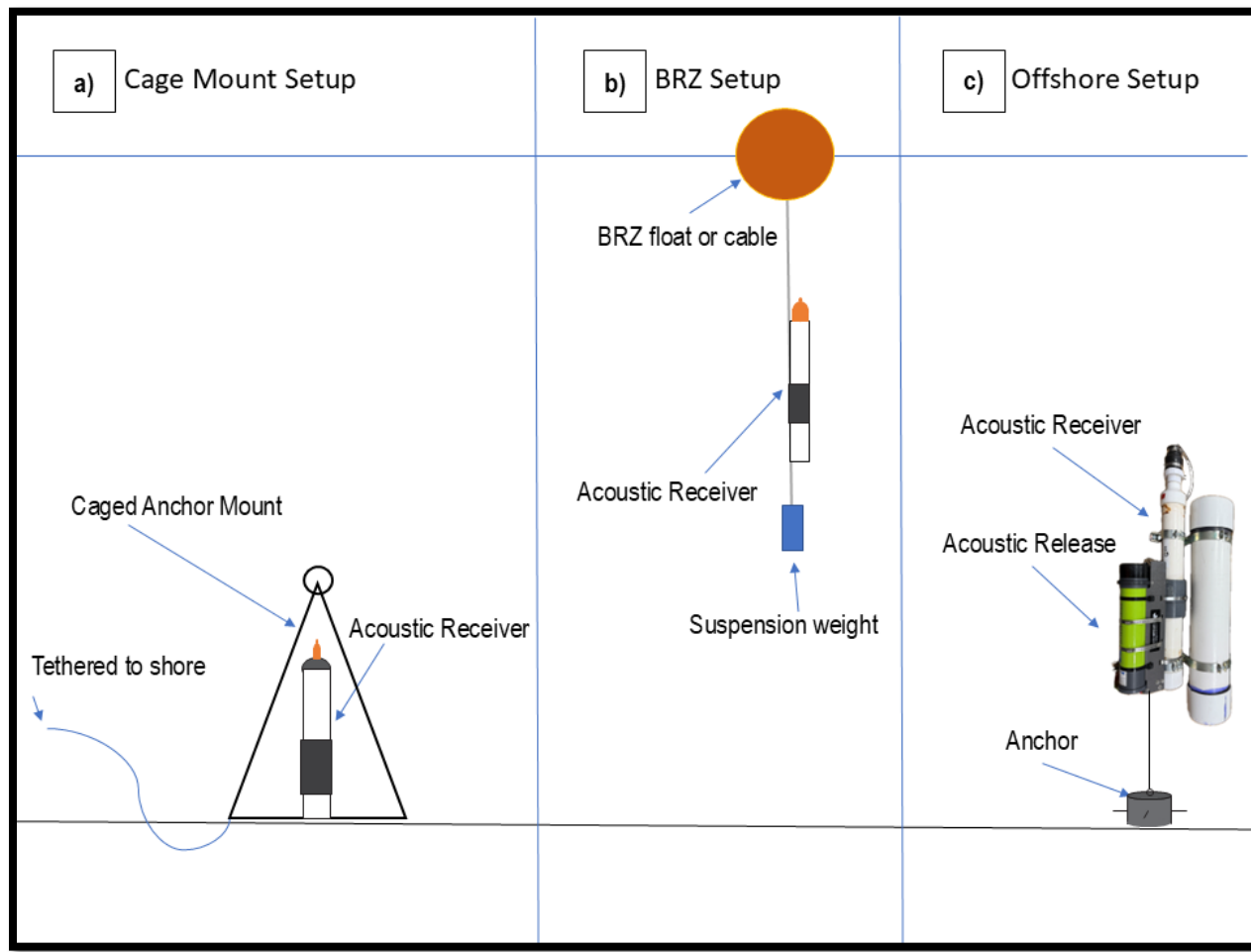


Figure 2. Schematics of the three different acoustic receiver deployment setups (Cage Mount, BRZ, Offshore) that will be used in the 2026 GPUD survival evaluation.

Grant PUD will prepare the acoustic receivers for deployment, which includes setting concrete anchors and assembling the necessary rigging and flotation components for each setup (Figure 2). LGL will lead the receiver deployment efforts, with support from Grant PUD personnel. Anchors will be loaded and deployed from a Grant PUD vessel at predetermined locations, while an LGL vessel will serve as a safety boat. Following each deployment, GPS coordinates will be recorded to document receiver positions. To assess receiver performance and detection range, each unit will be tested using GPS-tracked test tag drags. Throughout the study period, deployed receivers will be routinely checked for functionality and downloaded when conditions allow. At the end of the study period (until June 30) all the receivers will be recovered, and the detection data will be downloaded.



Figure 3. Maps of the array configurations at Crescent Bar (A), Sunland Estates (B), and Mattawa (C).

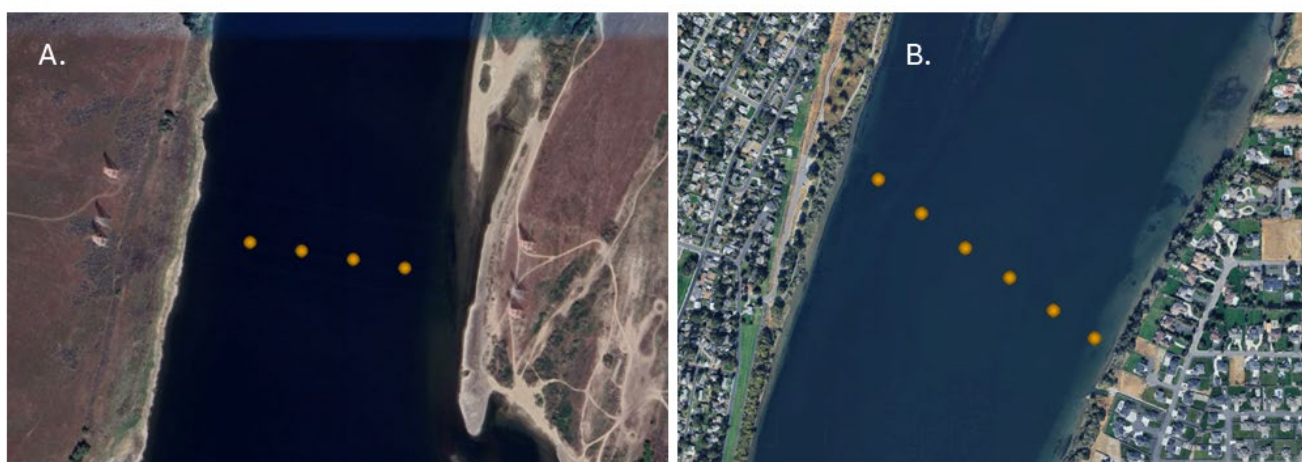


Figure 4. Maps of the terminal array configurations at Lower Ringold (A) and Hanford (B).

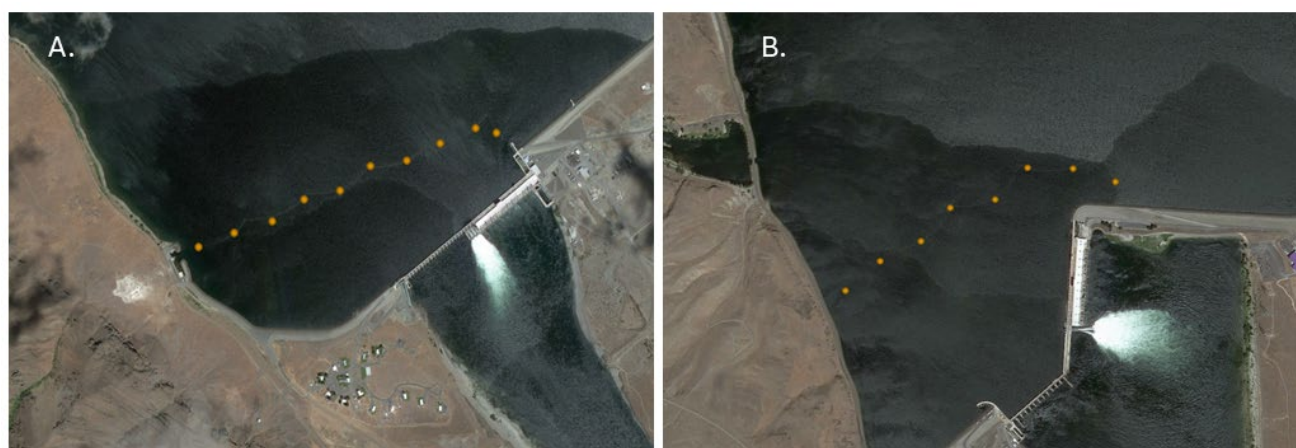


Figure 5. Maps of array configurations on the Boat Restricted Zone (BRZ) buoy lines at Wanapum (A) and Priest Rapids (B) dams.

Table 2. Quantities of acoustic receivers to be deployed separated by site name and deployment method.

	BRZ	Cage Mount	Offshore	Total
Crescent Bar	-	2	2	4
Sunland Estates	-	2	3	5
Wanapum Dam	10	-	-	10
Mattawa	-	2	3	5
Priest Rapids Dam	8	-	-	8
Lower Ringold	-	2	2	4
Hanford	-	1	5	6
Total Receivers	18	9	15	42

4.3 Fish Collection and Transport

Juvenile steelhead and yearling Chinook Salmon study fish for the 2026 survival evaluation will be collected from the Rock Island juvenile collection facility via an Interlocal Agreement Grant PUD has with the Public Utility District No. 1 of Chelan County, Washington. At the Rock Island juvenile facility, fish will be sorted by an LGL staff member in a light MS-222 solution by species, size, and physical condition.

Individual fish will be accepted for tagging based on pre-determined acceptance/exclusion criteria, as outlined by the Columbia Basin Surgical Protocol Steering Committee (2011) and the Tagging and Fish Husbandry QAQC Plan developed by LGL (Appendix B). These criteria are as follows:

Fish accepted if it:

- Is a yearling Chinook Salmon or juvenile steelhead;
- Fork length is between 95 – 210 mm;
- Has either an intact or clipped adipose fin;
- Is not previously tagged with a coded wire, elastomer, PIT, or acoustic tag; and
- It weighs enough that its tag burden will not exceed the maximum threshold percentage (3%).

Fish excluded if it:

- Is a non-target species;
- Is moribund or emaciated;
- Shows external signs of prior tagging (e.g., radio, acoustic,);
- Indicates a positive reading when read with a PIT-tag reader;
- Has malformations such as spinal deformities;
- Exhibits descaling >20% on any side of body;
- Exhibits any sign of gas bubble trauma (GBT) or bacterial kidney disease (BKD);
- Has physical injuries severe enough to impede performance, such as: opercula damage, exophthalmia, eye hemorrhages (>10% of eye), head or body injuries, or missing fins and/or extreme erosion; or
- Shows evidence of infections; symptoms include fungal infections on body surface, gill necrosis, open lesions on body or fins, swollen body, ulcers, or copepod parasites on the eyes or gills.

Fish that pass the sorting criteria will be held in sanctuary buckets segregated by species before being transferred to transport tanks on a Grant PUD transport truck in preparation for transport to the tagging facility at Wanapum Dam (Fishtown). The transport tanks will be kept on flow-through river water until transportation occurs.

Grant PUD will provide the necessary transportation equipment (e.g., fish transport vessels) and transport all study fish from the Rock Island juvenile collection facility to Wanapum Fishtown daily over the course of the evaluations. During transport, on the back of a flatbed truck, oxygen will be supplied into the transport tanks and monitored to maintain appropriate levels. The water temperature and dissolved oxygen (DO) within each transport tank will be recorded at the start and end of transportation.

Upon arrival at the tagging facility at Wanapum Dam, the sanctuary buckets will be removed from the transport vehicle and transferred to mobile 30 gallon holding vessels with flow-through river water supplied by water pumped from the fish ladder. To minimize the stress on study fish, transported fish will be held in the holding vessels for approximately 24 h before tagging occurs the following day. While being held in the holding vessels, personnel will measure DO and water temperature twice a day using a calibrated oxygen/temperature meter. The objective will be to maintain oxygen levels at 9-12 mg/L and water temperatures within 2°C of ambient river water. Fish will not be fed while being held.

Following arrival to Fishtown, fish handling, care, and the surgical implantation of tags will be managed by LGL.

4.4 Fish Tagging and Surgery

The surgical implantation procedures used will be based on the methods outlined in Adams et al. (1998) and Liedtke et al. (2012a,b) and those used during the acoustic-tag studies at Wanapum and Priest Rapids dams conducted in 2008-2011, 2014-2017 (e.g., Sullivan et al. 2009; Timko et al. 2010, 2011; Thompson et al. 2012; Hatch et al. 2015, 2017). See Appendix B for complete details regarding tagging procedures.

Immediately before surgery, fish will be transferred to an anesthetic bath containing tricaine methanesulfonate (MS-222, 60-80 mg/L) until equilibrium is lost. Anesthetized fish will be measured to the nearest millimeter and weighed to the nearest tenth of a gram and scanned for a PIT tag and CWT before being placed on a Plexiglas surgical table. The surgical table will be soaked with Stress Coat (*Aquarium Pharmaceuticals, Inc. Chalfont, PA*) to minimize scale loss and maintain the exterior mucous coat. Fish will be placed ventral side up on the surgical pad, and their gills will be continuously flushed with anesthetic solution fed through a tube placed in the mouth of the fish. Immediately prior to the end of surgery, the flow of anesthetic solution will be replaced with fresh river water to start the recovery process. River water supplied from within the fish ladder entrance will serve as the water source during tagging operations.

To implant the acoustic transmitter and PIT tag into the fish, an incision will be made 3 mm away from and parallel to the mid-ventral line starting about 3 mm anterior to the pelvic girdle. The incision will be only deep enough to penetrate the peritoneum (Adams et al. 1998). The PIT tag will be implanted first, followed by the acoustic transmitter which is implanted with the transducer pointing towards the head of the fish. Both tags will be positioned horizontally directly under the incision. The incision will be closed with a single absorbable suture placed across the incision. Surgical equipment will be disinfected with a diluted germicidal solution after each fish.

Immediately following surgery, tagged fish (2 fish per bucket) will be moved to 5-gal oxygenated recovery buckets where they will be monitored until they gain equilibrium and begin regular swimming and breathing movements. Recovery buckets will then be placed on shelves in a recovery room, adjacent to the tagging trailer. Each bucket will have an opening in the lid that allows for continuous water flow, and a PVC overflow spout. Dissolved oxygen and water temperature will be measured twice a day. The recovery room will be supplied with two redundant pumps in case of pipe blockage in one. Water flow to the recovery buckets will be checked on a routine schedule by LGL in coordination with Grant PUD staff. Checks will occur a minimum of 3 times per day.

Approximately 24 h after tagging, the tagged fish will be moved into fly tanks supplied with ambient river water. The yearling Chinook Salmon and juvenile steelhead will be released via these fly-tanks which are transported to the designated release locations (either Rock Island Dam tailrace or Priest Rapids Dam tailrace) via helicopter. Fish will be held in the fly tanks for several hours until their release into the river. When ambient river water is not being circulated through the fly tanks, supplemental oxygen will be provided. Grant PUD staff will assist LGL in the loading of buckets and be responsible for moving fly tanks to helipad.

4.5 Fish Releases

Study fish releases will be conducted via a helicopter departing from Wanapum Fishtown, as done in the 2014-2017 survival studies. Grant PUD will secure the necessary helicopter contractor to support the survival evaluations in 2026.

Approximately 2 h prior to departure from the Wanapum Dam holding site LGL will be responsible for measuring and recording DO and water temperature within the fly tanks. Measurements will be taken immediately before and immediately after the tagged fish are transferred into the fly tanks. Additionally, LGL will conduct a pre-release verification of the acoustic tagged study within the fly tanks that are prepared for release. The pre-release verification acts as a final QA/QC procedure to confirm the operability of the acoustic tags before the release commences. Any study fish not detected during the pre-release verification will be assumed to have a failed acoustic tag and will be censored from the survival evaluation.

For all releases, water flow to the fly tank will be shut off 10 minutes prior to departure by Grant PUD staff. An oxygen tank, attached to each fly tank will provide oxygenation to the water during transport. Fly tanks will be attached to a helicopter by longline cable and will be transported to their release site. The helicopter will lower the fly tank to within about 1 m of the water and will trigger a release mechanism that opens a valve near the bottom of the fly tank which allows the fish to pour out into the river. It will take approximately 60 seconds for the fly tank to be emptied into the river.

Every fish release will be monitored by onshore personnel who will communicate directly with the helicopter pilot to help facilitate a fish release that is near the surface of the water. Once a successful release occurs, the monitor will record the release time, approximate release height, as well as any additional comments. Releases below Rock Island Dam will be monitored by LGL staff while releases below Priest Rapids Dam will be monitored by Grant PUD staff.

4.6 Study Schedule

The study described above will be conducted in the spring season of 2026. It is anticipated that system deployment will be completed by April 15, 2026, and that the releases of tagged study fish will occur from approximately mid-April to June 1. Data collection will occur until June 30, at which time all deployed receivers will be recovered.

Each release group, including both treatment and control, will consist of 24 replicates as outlined in Table 3. To maintain consistency and reduce potential tagger bias, corresponding treatment and control replicates (e.g., RI01 and PR01, Table 3) will be tagged by the same individual. All study fish included in the 2026 evaluation will be tagged by one of three designated LGL personnel, identified as Taggers A, B, and C in Table 3. Each Rock Island release group (treatment) will include 41–42 juvenile steelhead and yearling Chinook, while each Priest Rapids release group (control) will include 33–34 fish. Additionally, every Priest Rapids release group will incorporate 2–3 dead juvenile steelhead and yearling Chinook, which will be released concurrently with the live fish.

Table 3. Mock-up release schedule for the 2026 juvenile steelhead and yearling Chinook Salmon survival evaluation through the Priest Rapids Project.

Dates	Juvenile Steelhead								Yearling Chinook							
	Rock Island				Priest Rapids Dam				Rock Island				Priest Rapids Dam			
Release Date	Tagger	Release	Live	Dead	Tagger	Release	Live	Dead	Tagger	Release	Live	Dead	Tagger	Release	Live	Dead
4/23/2026																
4/24/2026																
4/25/2026									A	RC01	41	0				
4/26/2026									A	RC02	41	0				
4/27/2026									A	RC03	41	0				
4/28/2026									A	RC04	41	0	A	PC01	33	2
4/29/2026									B	RC05	42	0	A	PC02	33	2
4/30/2026									B	RC06	42	0	A	PC03	33	2
5/1/2026									B	RC07	42	0	A	PC04	33	2
5/2/2026	A	RI01	41	0					B	RC08	42	0	B	PC05	33	2
5/3/2026	A	RI02	41	0					C	RC09	42	0	B	PC06	33	2
5/4/2026	A	RI03	41	0					C	RC10	42	0	B	PC07	33	2
5/5/2026	A	RI04	41	0	A	PR01	33	2	C	RC11	42	0	B	PC08	33	2
5/6/2026	B	RI05	42	0	A	PR02	33	2	C	RC12	42	0	C	PC09	34	2
5/7/2026	B	RI06	42	0	A	PR03	33	2	A	RC13	42	0	C	PC10	34	2
5/8/2026	B	RI07	42	0	A	PR04	33	2	A	RC14	42	0	C	PC11	34	2
5/9/2026	B	RI08	42	0	B	PR05	33	2	A	RC15	42	0	C	PC12	34	3
5/10/2026	C	RI09	42	0	B	PR06	33	2	A	RC16	42	0	A	PC13	34	3
5/11/2026	C	RI10	42	0	B	PR07	33	2	B	RC17	42	0	A	PC14	34	2
5/12/2026	C	RI11	42	0	B	PR08	33	2	B	RC18	42	0	A	PC15	34	2
5/13/2026	C	RI12	42	0	C	PR09	34	2	B	RC19	42	0	A	PC16	34	2
5/14/2026	A	RI13	42	0	C	PR10	34	2	B	RC20	42	0	B	PC17	33	2
5/15/2026	A	RI14	42	0	C	PR11	34	2	C	RC21	41	0	B	PC18	33	2
5/16/2026	A	RI15	42	0	C	PR12	34	3	C	RC22	41	0	B	PC19	33	2
5/17/2026	A	RI16	42	0	A	PR13	34	3	C	RC23	41	0	B	PC20	33	2
5/18/2026	B	RI17	42	0	A	PR14	34	2	C	RC24	41	0	C	PC21	33	2
5/19/2026	B	RI18	42	0	A	PR15	34	2					C	PC22	33	2
5/20/2026	B	RI19	42	0	A	PR16	34	2					C	PC23	33	2
5/21/2026	B	RI20	42	0	B	PR17	33	2					C	PC24	33	2
5/22/2026	C	RI21	41	0	B	PR18	33	2								
5/23/2026	C	RI22	41	0	B	PR19	33	2								
5/24/2026	C	RI23	41	0	B	PR20	33	2								
5/25/2026	C	RI24	41	0	C	PR21	33	2								
5/26/2026					C	PR22	33	2								
5/27/2026					C	PR23	33	2								
5/28/2026					C	PR24	33	2								
5/29/2026																
5/30/2026																
	Rock Island Steelhead 1000 0				Priest Rapids Steelhead 800 50				Rock Island Chinook 1000 0				Priest Rapids Chinook 800 50			

5.0 Analysis and Reporting

Upon completion of the study migration period, detection data will be downloaded from recovered receivers and then uploaded to the Grant PUD project SQL Server database. Once the data are on the SQL Server, duplicate or false records (i.e., acoustic noise) will be filtered using the methods outlined by Thompson et al. (2012) and the detection data will then be loaded into tables and evaluated for accuracy. Data validation and quality control will include multiple validation procedures based off the US Army Corps of Engineers protocols used ubiquitously in acoustic tag research. Quality control will also include evaluating normal downstream movement versus other movements (i.e., upstream movement).

5.1 Statistical Analysis

The statistical analysis for this study will be conducted by Buchanan Statistical Consulting. They will be responsible for providing integrated statistical support from study design through data analysis, and interpretation of results. These tasks will be carefully coordinated by all parties involved (i.e., Buchanan Statistical Consulting, LGL Limited, and Grant PUD staff).

The statistician will perform the survival analysis using a modified Cormack-Jolly-Seber analysis to estimate project survival using the acoustic-tag data (Cormack 1964, Jolly 1965, Seber 1965). Tests of model assumptions will be performed. Standard errors and confidence intervals will be provided for the estimated parameters. Estimates will be provided for each of the replicate release-pairs along with modeling to identify the most parsimonious (and most precise) model that adequately describes the data. Seasonal averages will be computed along with associated standard error. Tests of assumptions will be based on Burnham tests 1-3 (Burnham et al. 1987) along with Chi-square tests of mixing. Among-year comparisons of survival estimates will be performed.

For further details regarding the statistical analysis that will be used, please see Appendix A. Statistical Methods for J. Skalski's statistical study plan for the 2015 survival study – *Statistical Plan for the Sockeye Salmon and Steelhead Survival Studies at Wanapum and Priest Raids Projects in 2015*.

5.2 Reporting

A draft report, including introduction, study area, methods, and results, and charts of smolt counts for both accepted and rejected fish (indicating reason for rejection, adipose fin present or absence, and whether a CWT was present) is anticipated to be presented to Grant PUD by December 15, 2026. Once comments are received from Grant PUD and the Priest Rapids Coordinating Committee on the draft report, a final report will be provided to Grant PUD within 60 days.

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Appendix A. Statistical Methods

A1 Introduction

The acoustic-tag study at Grant PUD in 2026-28 will have the single objective of determining project survival for yearling Chinook, juvenile steelhead, and sockeye passing through the Priest Rapids Project, which includes Wanapum and Priest Rapids dams and reservoirs. This objective will be accomplished using a coordinated paired release–recapture study of acoustic-tagged yearling Chinook, juvenile steelhead, and sockeye above and below the Priest Rapids Project.

A2 Release-Recapture Design

The objective of the 2025-27 acoustic-tag survival study will be accomplished using two release locations, one in the Rock Island tailrace and the second in the Priest Rapids tailrace.

A2.1 Paired Release-Recapture Design

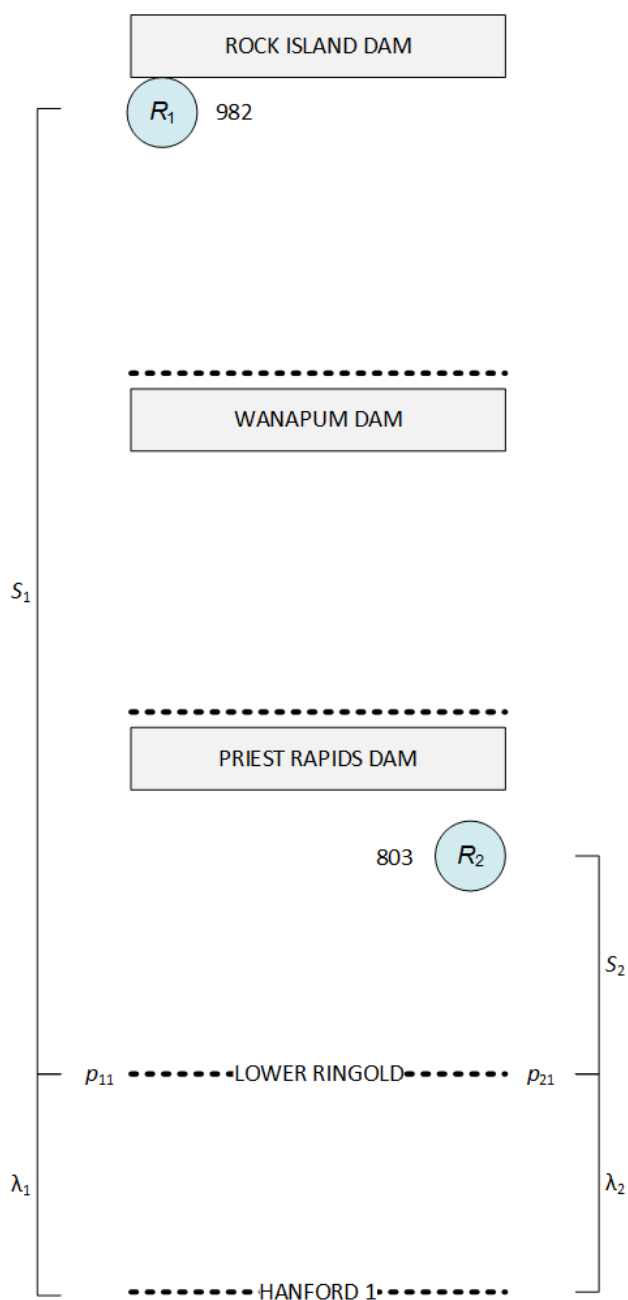
Project passage survival through the Priest Rapids Project will be estimated using the paired release-recapture design of Burnham et al. (1987). Releases from Rock Island (R_1) and Priest Rapids (R_2) tailraces will be used to estimate yearling Chinook and steelhead passage survival through the Priest Rapids Project (Figure A1). A total of 1,000 acoustic-tagged yearling Chinook, juvenile steelhead, and sockeye will be released at the Rock Island tailrace (R_1) and 800 at the Priest Rapids tailrace (R_2). Detection arrays at Lower Ringold and Hanford 1 will be used as the downstream detection locations (Figure A1).

A2.2 Tag-Life Study

A total of 50 tags for each the yearling Chinook, juvenile steelhead, and sockeye will be systematically sampled from the single tag lot for evaluating the tag-life. The tags will be activated and monitored continuously in ambient river water until all tags had failed. The failure times will be analyzed to produce a tag-life curve that will be used to produce tag-life-corrected survival estimates. The data will be fit to the vitality curve of Li and Anderson (2009).

A2.3 Dead Tagged Fish Releases

To test for the possibility of “false/positive” detections of dead fish by the downstream detection arrays, 50 (dead) smolts of each species will be implanted with an acoustic tag and release into the Priest Rapids Dam tailrace, along with the control group of tagged smolts.



Priest Rapids Project Passage Survival: $\hat{S} = \frac{\hat{S}_1}{\hat{S}_2}$

Figure A1. Schematic of the paired release-recapture design to be used to estimate project passage survival through the Priest Rapids Project. Release sizes will be 1,000 (RI) and 800 (PRD).

A3 Statistical Analyses

A3.1 Tests of Assumptions

Burnham et al. (1987) Tests. Tests 2 and 3 of Burnham et al. (1987) will be used to assess whether upstream detection influences downstream survival. These tests are most appropriate when fish are physically recaptured or segregated during capture. However, acoustic-tag studies do not use physical recaptures to detect fish. Consequently, active-tag studies rarely, if ever, result in significant tests T2 and T3 beyond expected α levels of occurrence. Furthermore, the very high to nearly complete detection probabilities result in data often too sparse to analyze using Burnham tests 2 and 3. For this reason, they were not scheduled to be used.

Tests of Mixing. Chi-square tests of homogeneity will be used to test whether release groups R_1 and R_2 were mixed upon arrival at downriver detection sites. A contingency table test of homogeneous arrivals over time will be constructed of the form:

	R_1	R_2
1		
2		
3		
$\vdots$		
$\vdots$		
D		

Typically, such chi-square tests are very sensitive to departures from homogeneous arrival distributions and are most often significant, even in the best of circumstances. Graphs of cumulative arrival distributions will be more useful in detecting systematic and meaningful departures from mixing.

Test of Tagger Effects. The effects of tagging and handling smolts by different taggers may not be manifested until sometime after release. To avoid biasing studies, tagging efforts by the tagging crews must be homogeneous across release groups used in the paired-release studies. Two different analyses will be used to help assure differential post-release handling mortality is not affecting study results.

First, analyses will be performed to evaluate whether the proportions of fish tagged by different taggers are similar across release groups. Chi-square $R \times C$ contingency table tests will be used to test for significant departures from homogeneity. Second, single release-recapture models will be run on the fish from different taggers to determine whether reach survivals is being affected.

An F-test will be used to test for homogeneous survival of the form:

$$F_{k-1,\infty} = \frac{S_{\hat{S}_i}^2}{\left[\frac{\sum_{i=1}^k \widehat{\text{Var}}(\hat{S}_i | S_i)}{k} \right]}$$

where

$$S_{\hat{S}_i}^2 = \frac{\sum_{i=1}^k (\hat{S}_i - \hat{\bar{S}})^2}{(k - 1)},$$

k = number of replicate survival estimates,

$\widehat{\text{Var}}(\hat{S}_i | S_i)$ = estimated sampling variance for the i th estimate of survival ($i = 1, \dots, k$).

A3.2 Tag-Life Curves

The tag-life data will be analyzed using the vitality curve of Li and Anderson (2009). The fitted probability density function will be used in Program ATLAS (<http://www.cbr.washington.edu/analysis/apps/atlas/>) to provide tag-life-adjusted survival estimates. One tag-life curve will be used for the Rock Island and Priest Rapids tailrace releases because as the tags will be activated at the same time and divided between the release sites for each replicate.

A3.3 Paired Release-Recapture Estimates of Survival

The paired release-recapture design of Burnham et al. (1987) will be used to estimate Priest Rapids Project passage survival based on tag-life adjustments of Townsend et al. (2006). An estimate of passage survival will be calculated for the Priest Rapids Project.

Estimation of Priest Rapids Project passage survival will be based on the R_1 and R_2 tailrace releases from Rock Island and Priest Rapids dams (Figure A1). Each release will produce four observable capture histories with probabilities of occurrence as denoted in Table A1.

Maximum likelihood estimation will be used to estimate model parameters (Table A1). A fully parameterized model will be used to estimate the parameters of the paired releases unless required precision is not achieved ($SE \leq 0.025$). In which case, likelihood ratio tests and AIC will be used to find the parsimonious model that best describes the data. Skalski et al. (2013) found fully parameterized models were the best balance between precision and bias when detection probabilities are very high, typical of acoustic-tag studies.

Table A1. Detection histories and expected probabilities of occurrence for release R_1 and R_2 from the acoustic-tag study at the Priest Rapids Project denoting survival (S), detection probabilities (p), tag-life corrections (L), and last reach joint survival and capture probability (λ).

Release	Detection History	Expected Probabilities
R_1	11	$S_1 p_1 L_{12} \lambda_1$
	01	$S_1 (1 - p_1) L_{12} \lambda_1$
	10	$S_1 p_1 (L_{11} - L_{12} \lambda_1)$
	00	$(1 - S_1) + S_1 [(1 - L_{11}) + L_{11} (1 - p_1) - L_{12} (1 - p_1) \lambda_1]$
R_2	11	$S_2 p_2 L_{22} \lambda_2$
	01	$S_2 (1 - p_2) L_{22} \lambda_2$
	10	$S_2 p_2 (L_{21} - L_{22} \lambda_2)$

$$00 \quad (1 - S_2) + S_2[(1 - L_{21}) + L_{21}(1 - p_2) - L_{22}(1 - p_2)\lambda_2]$$

The estimates of the survival and capture parameters in the likelihood the model were initially calculated, treating the estimates of tag life (i.e., $\hat{L}_{11}$, $\hat{L}_{12}$, $\hat{L}_{21}$, and $\hat{L}_{22}$) as known constants. Priest Rapids Project passage survival will be estimated as the quotient

$$\hat{S} = \frac{\hat{S}_1}{\hat{S}_2}. \quad (1)$$

However, to calculate a realistic variance estimator for the survival parameters, the error in the estimation of the tag-life probabilities must be incorporated into an overall variance calculation. The variance of the survival estimates will be calculated using the total variance formula

$$\text{Var}(\hat{S}) = \text{Var}_{\hat{L}}[E(\hat{S}|\hat{L})] + E_{\hat{L}}[\text{Var}(\hat{S}|\hat{L})]. \quad (2)$$

The above variance can therefore be estimated in stages using the expression

$$\text{Var}(\hat{S}) = s_{\hat{S}|\hat{L}}^2 + \text{Var}(\hat{S}|\hat{L}) \quad (3)$$

The second term in Equation (3) will be derived from the maximum likelihood model, conditioning on the tag-life probabilities (i.e., $\hat{L}$). The first variance component in Equation (3) will be calculated using bootstrap resampling techniques (Efron and Tibshirani 1993). Alternative estimates of L will be computed by bootstrapping both the observed tag-life data and travel-time data. For each estimated vector of tag-life parameters, survival will be estimated using a likelihood model. One thousand bootstrap estimates of the tag-life parameters will be calculated along with the corresponding conditional maximum likelihood estimates of survival. The first variance component in Equation (3) will then be estimated by the quantity

$$s_{\hat{S}|\hat{L}}^2 = \frac{\sum_{b=1}^{1000} (\hat{S}_b - \hat{\hat{S}})^2}{(1000-1)}$$

where

$\hat{S}_b$ = the b th bootstrap estimate of survival ($b = 1, \dots, 1000$);

$$\hat{\hat{S}} = \frac{\sum_{b=1}^{1000} \hat{S}_b}{1000}.$$

Use of Equations (2) and (3) will also permit examining the contribution of the sampling error in the tag-life parameters to the overall variance in survival estimates.

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Appendix B. Tagging and Fish Husbandry Plan

B1 Introduction

This document details the acoustic tagging and fish husbandry approach for the 2026-28 salmonid survival evaluations through the Priest Rapids Project. Specifics include the surgical tagging and suture methodology, duration and location of fish holding, transport methodology, maximum tag burden, minimum fish health criteria, husbandry methods (e.g., use of sanctuary nets, monitoring of dissolved oxygen and temperature, etc.), and details of accepted surgical procedures (e.g., anesthesia durations and concentrations, incision location and length, tag insertion and suture techniques). Tagging procedures and guidelines have been refined over more than a decade of studies (e.g., Timko et al. 2007a,b, 2010, 2011, Sullivan et al. 2008, Hatch et al. 2015, 2016, 2017, 2018, 2022) in order to maintain the highest standards of fish care.

B1.1 Purpose and Applicability

Procedures and guidelines for fish collection, handling, holding, and transport are included to reduce stressors to fish and increase the likelihood of a positive surgical outcome.

B1.2 Guiding Principles

Each individual tagged fish is a significant investment; therefore, efforts are made to ensure that the potential for handling and transmitter effects are controlled, and that tagged fish can reliably represent the untagged population.

B2 Collection, Sorting and Selection Criteria

B2.1 Collection

All study fish will be collected at the Rock Island Dam smolt monitoring facility. Fish that are destined for Wanapum Fishtown from the Rock Island smolt facility will be sorted and held for potential use in the survival evaluations; the selection criteria for fish size and condition are detailed below. Once the study begins, fish collection will need to occur daily to satisfy the tagging requirements of the study. On any given day, fish for treatment and control groups will be collected. To allow for secondary culling by surgeons for health criteria (detailed below), ~15% more study fish will be collected than are required to meet release targets. Selected fish will be held outside in a sanctuary box for transporting to the Wanapum Fishtown tagging facility. River water that is pumped through a nitrogen degassing tower will continually flow through the holding tanks and perforated buckets.

B2.2 Fish Size Criteria

- Tagged study fish will not exceed a 3% tag burden threshold.
- Study fish weighing at least 10.5 g will be double tagged with an acoustic transmitter and PIT (0.316 g, combined weight at 0.210 g and 0.106 g respectively).
- The minimum acceptable fork length is 95 mm due to elongated shape of the tag and the maximum allowable fork length is 210 mm to avoid tagging large fish that are more likely to revisualize.

B2.3 Fish Health Criteria

During the sorting process at the Rock Island facility, fish will be selected based on criteria including descaling, fungus, disease, prior markings or tags, and other criteria outlined in List II 'Culling Guidelines for Salmonid Smolts'.

B3 Transportation and Holding Schedule

Fish will be loaded and transported by Grant PUD crews on the same day as collection.

Following the recommendations of Liedtke et al. (2012), the total time that any given fish will be delayed (from time of capture to the time of release) will be approximately 48 hr to minimize stresses related to migratory delays (Marshall et al. 2011).

B4 Tag Site Fish Holding and Husbandry

B4.1 Pre-tag Holding

Following transport to the tag site holding facility, study fish will be supplied with flowing, untreated river water that has been passed through de-nitrogen tower.

B4.2 Tagging Site Pre-release

In the morning before release, final water quality measurements of dissolved oxygen and temperature will be recorded while the final tag codes are being checked. Once the final husbandry data are collected, custody of the fish will be given to Grant PUD staff, who will transfer the fish to the release fly boxes for transport via helicopter to the in-river release locations.

B5 Tag Plan

B5.1 Tag Randomization

Tags will be received from ATS in one batch no later than 1 April of each study year and will have all been manufactured by a single technician. Regardless of this uniformity in manufacturing, tags will be selected randomly for all upstream, downstream, and Tag Life replicates.

B5.2 Tag Activation

Each replicate will have randomly selected tags with associated tag codes which are activated in the afternoon or evening prior to the day of tagging. Programming will occur on site at the tagging location. A tag sheet with tag code assignments will be available to the technician performing tag activation. An ATS model Micro Sonic Pinger Dish IV will be used to activate, verify function, and confirm the code for every tag. Following activation, tags will be placed in tag boxes with separate and uniquely identified slots for each tag and placed in a disinfection bath for a minimum of overnight (details below) until tagging occurs on subsequent days.

B5.3 Pre-release Verification

Each morning prior to release, a Model SR3017 acoustic receiver will be used to verify the expected operation of each implanted tag in a replicate release. The hydrophone will be placed in each bucket and detections monitored to verified tag functionality before release.

B6 Surgical Procedures

B6.1 Setup of Equipment

See equipment listed in List I.

- A. All tags will be disinfected in diluted Chlorhexidine solution for a minimum of 10 minutes. The tags will then be transferred to a distilled water rinse bath, then actively rinsed under running distilled water. The tag will be

placed, using forceps, in a vessel of distilled water just prior to implantation. Note that certain disinfectants may adversely affect the coating on the transmitters.

- B. Surgical tables (with embedded measuring board) and surgical equipment will be prepared for use. All staff must be trained on the use of all length measuring equipment. Approximately 1-2 ml of undiluted Stress Coat will be sprayed on the surgical table.
- C. The weigh-scale will be set up. All staff must be trained on the use of all weight measuring equipment.
 - i. Ensure the scale is functioning properly. Scales should be calibrated at the start of the season and whenever they are moved and should be recalibrated as necessary.
- D. The recovery buckets will be filled with untreated river water and supplied with oxygen if necessary.
- E. *Administration of anesthetic:* The effectiveness of MS-222 as an anesthetic varies with factors such as temperature, size of fish, species, exposure, and fish density. Adjustment of the anesthesia concentration will be based on the amount of time it takes for a fish to lose equilibrium. The surgeon will target between 60-80 mg/L MS-222 concentration level for salmonids. Communication among the tagging crew is very important to ensure that MS-222 is only administered to the anesthetic buckets and gravity feed buckets once. This will ensure that MS-222 is never administered into a bucket until it has been confirmed that no one else has already done so.
 - i. Anesthetic and freshwater buckets will be filled and prepared just prior to tagging.
 - a. The "HEAVY" anesthetic bucket will be filled with 10 L of untreated river water. We will start by adding approximately 10 ml of MS-222 stock solution to yield a concentration of 53 mg/L. We will adjust the amount of anesthesia concentration accordingly based on the amount of time it takes for a fish to lose equilibrium and the amount of water in the bucket.
 - b. Both gravity-feed buckets will be filled with 10 L of untreated river water. Add 3 ml of MS-222 stock solution to the bucket marked "MS-222" for a light dose. NO MS-222 will be added to the bucket marked "fresh water".
 - ii. Note that MS-222 is acidic and changes the pH level of water, therefore it is desirable to buffer the water by adding sodium bicarbonate. We will want to add the same quantity of buffer as MS-222. The easiest way to achieve this is by making the sodium bicarbonate stock solution at the same concentration as the MS-222 solution, so that an identical number of milliliters of each solution can be added to the buckets. For example, if 3 ml of MS-222 stock solution is added to a bucket, then 3 ml of sodium bicarbonate stock solution must also be added. The MS-222 stock solution will have been prepared by adding 1 kg of MS-222 powder to 5 gallons of water, yielding a stock solution of 52.63 mg/ml, or 53 g/L. Hence, the buffer stock solution will be made by mixing 53 g of sodium bicarbonate into 1 L of river water.
 - iii. Stress Coat helps maintain the slime coat on the fish, which helps prevent infection and reduce stress. We will mix a Stress Coat stock solution (25% Stress Coat, 75% river water), and add some

to all buckets (anesthetic, anesthetic gravity feed, and recovery buckets). For every 10 L of water, 10 ml of the Stress Coat stock solution will be mixed in.

- iv. Water in all buckets (anesthetic and gravity feed) will be monitored continuously and changed periodically to minimize dilution of anesthetic water and temperature changes (such as a temperature change of more than 2°C above or below ambient water temperature) and to ensure that there is sufficient water in your gravity feed buckets to last the complete duration of a surgical tag insertion. Typically, water in the anesthetic and gravity feed buckets will be discarded and refilled after approximately twenty fish are tagged. During warm days, water may be changed more frequently due to rapid increases in water temperature.
- v. All gravity-feed “MS-222” and “fresh water” buckets will be monitored for dissolved oxygen. Oxygen levels should be maintained near saturation. It will be the surgeon’s call to add oxygen to the necessary buckets.
- vi. Sanctuary nets will be used for the transfer of fish between pre-tag holding tanks and “HEAVY” anesthetic buckets.

B6.2 Water Quality During Surgery

All staff must be trained on all water quality measuring equipment.

- A. Dissolved oxygen (D.O.): will be measured in mg/L in all rectangular totes while any pre- or post-tag fish are being held.
 - i. Measurements will be taken using a D.O. meter.
 - ii. D.O. concentrations should be between 8 mg/L and 13 mg/L. If readings are outside of this range, the field manager will be informed, and totes will be checked for limited water inflow (which may cause low oxygen levels). Supplemental oxygen or increased flow will be used if necessary.
- B. Water temperature will be measured in °C in all rectangular totes while any pre- or post-tag fish are being held. Temperature may be taken with a D.O. meter if the tool can measure both. Temperature should be within 2°C of the ambient water temperature. If readings are outside this range, the field manager will be informed, and the water will be changed.
- C. Air temperature within the surgical area should be controlled to ensure that the fish on the surgical table are not too hot or too cold.

B6.3 Implantation of Tags

- A. Rejection of fish by surgeon
 - i. Each fish will be dip-netted (using a sanctuary net) from the large holding container and placed in the “HEAVY” anesthetic bucket. The lid of the anesthetic bucket will be secured as soon as the fish is in the bucket. After one minute, the fish will have slowed down enough to quickly check for abnormalities (visual check only while the fish remains unhandled in the sedative). If the fish is

rejected after one minute, it is then transferred to a freshwater recovery bucket and released into the river.

- ii. If the fish is not rejected after one minute, it will remain in the “HEAVY” anesthetic bucket for an additional 1-3 minutes until it has lost equilibrium, at which time it will be ready for tagging. If after sedating a few fish, it is observed that the time required for fish to lose equilibrium is more or less than normal, then the anesthetic concentration will be adjusted upwards or downwards. If the fish loses equilibrium in less than one minute or more than 5 minutes, the fish will be rejected. The data recorder will document the time the fish entered “HEAVY” to monitor the duration that each fish is in “HEAVY” anesthetic.
- iii. Once the fish loses equilibrium, the surgeon will visually screen the fish for tags, fin clips, fungus, disease, GBT, descaling, bloated belly, or any obvious abnormalities. Any necessary information will be relayed to the data recorder. See culling procedures for the type of fish that are acceptable for tagging (List II). If the fish is unacceptable for tagging, it will be transferred to a freshwater tank and then released into the river once it has regained equilibrium.

B. Recording fish length and weight

- i. Anesthetized fish will be transferred (using sanctuary net) to the scale and weighed to the nearest 0.1 g. The data recorder will stop the “HEAVY” sedation timing once the fish is removed from the “HEAVY” bucket. The data recorder will also document the start time of surgery at this point.
- ii. The fish will then be transferred to the measuring board (on the surgery table) to measure the fork length to the nearest millimeter. Data will be vocally relayed to the data recorder to avoid data errors. The data recorder will then record this information and repeat numbers back to avoid any miscommunication.
- iii. Any fish dropped on the floor or ground before tagging will be rejected. A fish that flops from the tagging trough into the surgical table during tagging can still be tagged, unless MS-222 water enters the incision. If a fish is dropped on the floor after it is tagged, the tag will be removed, and the fish will be rejected.

C. Surgery

- i. Fish will be placed on the surgery table ventral side up. A light anesthetic solution will be administered through the gravity feed tubing as soon as the fish is on the surgery table. The tubing will be placed just inside the mouth or as close as possible, so the water flows across the gills. The flow of water should be just enough to cover the gills. If the flow is too low, the fish will flare its gills and become agitated. The flow will be adjusted so that respiration of the fish remains normal throughout the surgery. The in-line valves will be used to control the flow of anesthetic, fresh water, or a mixture of both.
- ii. *Incision:* Using a scalpel, an incision of approximately 4 mm (dependent on tag size) will be made in a position about 3 mm away from and parallel to the mid-ventral line. The incision will start a few millimeters anterior to the pelvic girdle (Figure B16). The incision will be just deep enough to penetrate the peritoneum (the thin membrane separating the gut cavity from the musculature),

avoiding the internal organs. The spleen is close to the incision position, so it is important to pay close attention to the depth of the incision.

- a. There is no exact specification for what size scalpel blade to use for each fish. For most smolts, a Beaver micro-unitome knife, 15 degrees 5 mm disposable scalpel will be used.
- b. One scalpel blade can be used on about 7-10 fish before it becomes dull. If the blade is pulling roughly or making jagged incisions, it will be changed immediately.
- c. If the surgeon believes that an internal organ was cut, they will not implant the tag. Instead, they will stitch the incision and reject that fish. Excessive bleeding will be noted on the datasheet.

iii. *Insert tag:* Once the incision is open, the PIT tag will be inserted first followed by the acoustic tag for double tagged fish. The acoustic transmitter end will be inserted first through the incision and should be pointing to the head of the fish; the battery end of the tag should then be pointing posteriorly (i.e., toward the tail; see Figure B27). The tag should lie directly under the incision and will aid in keeping the organs from protruding through the incision. If the tag is dropped prior to being implanted into the fish, a different tag will be inserted (it will be necessary to soak the dropped tag in a Chlorhexidine solution for a minimum of 10 minutes, then rinsed thoroughly with distilled water before being put into another fish).

iv. *Suturing:* Two interrupted stitches are usually used to close the incision, but the number depends on the size of the tag and incision.

- a. To make a stitch, the needle should be locked (at the end of the suture) in the needle drivers so the needle point faces the surgeon. The needle will be inserted on the far edge of the incision and will exit on the other side the incision (the side closer to the surgeon), and the suture will be pulled perpendicular through the two edges. The needle will enter and exit the skin as close to the edge of the incision as possible without tearing the skin (~ 2 mm from edge of incision). The needle and suture will be pulled through the skin to leave a tag end of about 2-3 cm of suture material protruding from the needle entrance location and will then be released from the needle drivers. With the surgeon's non-dominant hand, they will grasp the long end of the suture material (usually with thumb and forefinger) at or below the needle, and make two forward wraps (i.e., away from the surgeon's body) around the tip of the needle driver, which should be held in the surgeon's dominant hand. With the two wraps still around the needle driver, the short end of suture material will be grasped with the needle driver, and the stitch will be tightened by pulling the wraps off the needle driver and pulling both ends of suture material perpendicular to the incision. On the first knot, with the dominant hand holding the needle driver, the surgeon will pull toward their body, with the non-dominant hand pulling away from their body. The suture will be tightened lightly, just so the edges of the incision meet but do not overlap, pucker, or bulge the edges of the incision. The second knot will be in reverse order from the first knot, and there will be only one wrap with the suture material around the needle holder. The second knot will be looser than the first, again with care taken to not to overlap, pucker, or bulge the edges of the incision (this allows room for tissue swelling). On the third knot, the long end of suture material will be grasped with the surgeon's non-dominant hand, one reverse wrap will be made (i.e., towards the surgeon's body) around the end of the needle driver, and the short end of suture

will be grasped with the needle driver to tighten the stitch. This time, the knot will be tightened by pulling the dominant hand (holding the needle drivers) away from the surgeon's body and the non-dominant hand toward them. This completes one stitch. The suture will be cut with the needle drivers, leaving ends approximately 3 mm in length.

- b. When making wraps around the needle driver, it is easiest to make the wraps closer to the hand holding the suture, rather than closer to the fish. After making wraps around the needle driver, the short end of the suture will be grabbed (with the needle drivers) close to the end of the suture material. If the suture material is grabbed closer to the fish (instead of closer to the end of the suture), then the tag end will fold onto itself, get tightened into the knot, and leave two strands of suture material in the knot. This is a common mistake.
- c. When pulling a knot tight, we will be sure that the knot lays flat and does not twist onto itself into a "balled-up" knot. If the knot begins to "ball-up" when pulling tight, it can sometimes be coaxed to lay flat by twisting the suture material between a thumb and forefinger. With experience, each surgeon develops their own "tricks" for making sure the knot lays flat. Although not every knot is perfect, we will strive for perfection on every stitch, rather than settling for an imperfect stitch. An imperfect stitch will be more likely to come untied, possibly resulting in slower wound healing or tag loss, which could ultimately affect the survival of the fish or bias the study results.
- d. There is no exact specification for what size suture to use. 5-0 sutures can be used for most salmon smolts. A tapered needle is typically used on all juvenile salmonids. Since fish come in all sizes, there will be some overlap in these approximate parameters.
- e. Generally, a good time to switch the in-line valve on the gravity feed buckets or hose to untreated river water is just prior to stitching. This initiates recovery from anesthesia as early as possible. However, if the fish appears to be inadequately gilling, we will provide a mixture or all fresh water as soon as possible. If the fish is reviving too quickly and the surgery is not yet complete, we will not switch to fresh water.
- f. If the incision is too long, it is acceptable to add a second stitch. This information will be relayed to the data recorder so that note can be made on the datasheet.
- g. Because suture material is long, each individual suture (one packet) can be used on 8-12 fish.
- v. When multiple successive fish are to be tagged, and to keep the tagging process flowing, the anesthesiologist will start sedating the next fish before the first fish's surgical procedure is complete. The surgeon should indicate to the anesthesiologist when the next fish should be sedated. Typically, a good time to start sedation on the next fish is immediately after the previous fish has been taken out of the "HEAVY" bucket. Timing may have to be adjusted based on surgery time.

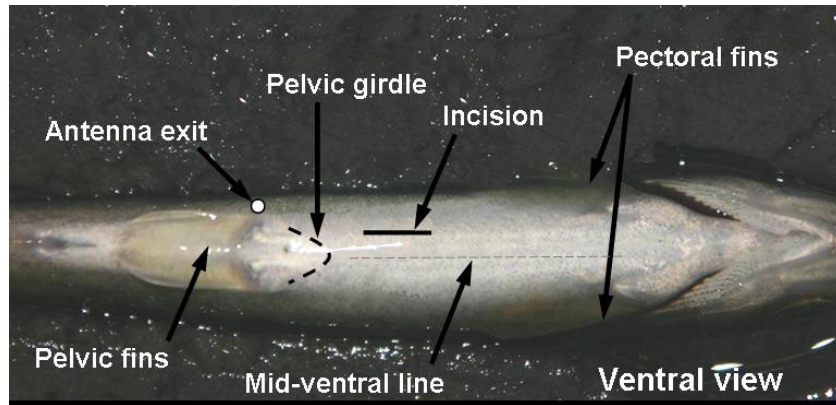


Figure B16. Ventral view of a juvenile salmon on a tagging platform. Locations of the pelvic girdle, incision and antenna exit site (for radio tags only) are shown.



Figure B27. Ventral view of a juvenile salmon showing insertion of acoustic tag into body cavity.

B6.4 Post-Surgery

- A. Fish will be transferred from the surgery table directly to a labeled 20 L recovery bucket using gloved hands. The data recorder will document the end time of surgery at this point. The data recorder will also record the label of the recovery bucket for that fish. There should be no more than 2 fish per recovery bucket. Dissolved oxygen levels should be consistent with that of the river as the recovery buckets are filled with water just prior to the fish being placed in the bucket. If there is a delay before a recovery bucket is used (no fish are in the bucket) due to rejecting fish, it is the surgeon's decision whether to add oxygen to the recovery bucket. Super saturation of oxygen has been proven to aid in recovery, so it is recommended to add supplemental oxygen to the recovery bucket once the first fish is placed in it. Supplemental oxygen will be removed once the second fish is put into the recovery bucket, and the bucket will be moved to the bucket rack to be placed on flow through river water. The bucket will remain super saturated for a few minutes until all water is flushed, allowing the second fish to also benefit from super saturation during recovery. If several culls occur between the surgeries of the first and second fish, we will remove the supplemental oxygen from the bucket to avoid a prolonged period of super saturation.

- B. Between surgeries, the surgeon will prepare their tools for the next surgery. The tools will be disinfected in Chlorhexidine solution. We will ensure that the scalpel blade and suture are acceptable to use on the next fish, and if not, they will instead be replaced. All surgical equipment will be disinfected (by soaking them in Chlorhexidine solution for several minutes – see directions on bottle, typically at least 10 minutes) between each fish to avoid transmitting disease among individuals. Once disinfected, the tools will be rinsed thoroughly with distilled water. It may be necessary to have multiple sets of surgical instruments which are rotated in order to ensure the minimum recommended soak time between uses. Organic debris in the disinfectant bath reduces its effectiveness, so the disinfectant baths will be changed (refreshed) regularly.
- C. Tagged fish will be checked at various times throughout the day. Typically tagged fish will be checked approximately 30 minutes after surgery. If a tagged fish does not exhibit signs of proper swimming behavior or has not gained equilibrium it is up to the surgeon to decide if the fish should be rejected. Manual ventilation can be attempted by holding the fish in an upright swimming position in the water and moving the fish back and forth to force water flow through the gills. If the fish is rejected then the tag is pulled, soaked in a Chlorhexidine solution for a minimum of 10 minutes, then rinsed thoroughly with distilled water before being put into another fish. The rejected fish is re-sutured and released into the river.
- D. At the end of a tagging session, and after all the fish have been checked and re-checked, the fish will be moved from the 20 L recovery buckets into the 5-gallon post-tag holding containers, with a fish density of 2 fish per bucket.

B6.5 Post-Operative Monitoring

- A. D.O. and temperature will be measured hourly, as described above. Post-surgery fish health will be checked twice before the taggers leave for the day.
- B. Optimal post-op recovery time will be 16-22 hours and will not exceed 36 hours. The first fish tagged in a day could be held about 8 hours longer than the last fish of the day. All fish held at the tagging facilities, including post-operative tagged study fish, are monitored.

B6.6 Clean up

At the end of the tagging day:

- A. All counter tops, scales, and measuring boards will be wiped down with the Germiphene solution to disinfect. Germiphene solution should not be made of river water, but rather distilled water.
- B. Scalpels, forceps, and scissors will be soaked in Chlorhexidine solution, and then rinsed with distilled water and thoroughly dried to prevent rusting.
- C. Needle drivers will be scrubbed with a small brush or scour sponge.
- D. All autoclavable surgical equipment will be autoclaved prior to the next day's tagging session.
- E. Buckets and nets will be washed with Germiphene solution and then rinsed thoroughly with untreated river water and placed upside down to dry. Electronics will be stored in proper cases.
- F. Extra fish (not required for backup) in the pre-tag holding tanks will be released back into the river. Holding tanks will be emptied, flushed, and re-filled (for use the next day), or left empty.

B6.7 Data/Datasheets

The appropriate tagging and water quality data sheets will be filled out before, during, and after tagging. At the end of the tagging day, the field manager will review these data/datasheets to ensure proper collection procedures were followed.

B7 Release Procedures

As indicated above, the morning before release, final water quality measurements of dissolved oxygen and temperature will be recorded while the final tag codes are being checked. Once the final husbandry data are collected, custody of the fish will be given to Grant PUD staff, who will transfer the fish to the release fly box for transport via helicopter to the in-river release site.

The following details (where, when, and how) of the release procedures will be developed by Grant PUD in consultation with the Blue Leaf and LGL husbandry experts and will be documented elsewhere.

- Methods for transport of fish via helicopter to the release site
- Physical release locations
- Timing of release events
- Handling protocols to release of fish via helicopter into the river

B8 Surgical Equipment Materials List

- Safety vests, safety glasses, hard hats (**Grant PUD provides**)
- Dissolved oxygen (D.O.) meters which include temperature readings (**Grant PUD provides**)
- pH meter (**GCPUD Provides**)
- Total dissolved gas (TDG) meter (**GCPUD Provides**)
- Tags, acoustic (ATS400) (**Purchased and provided by Grant PUD**)
- Tricaine methanesulfonate (MS-222; 53 g/L stock solution) (**Grant PUD provides**)
- 20 L buckets marked at 10 L and clearly labeled “HEAVY” (referring to anesthetic buckets) (**Grant PUD provides**)
- 20 L numbered recovery buckets with lids (**Grant PUD provides**)
- Pair of 20 L buckets (gravity feed) marked at 10 L connected by rubber tubing with in-line shut-off valves – one labeled “MS-222” and one labeled “fresh water” (**Grant PUD provides**)
- Oxygen delivery system and oxygen supply– (**Grant PUD provides**)
- Waterproof scale measuring to the nearest 0.5 g (**Grant PUD provides**)
- Surgery table – plexi-glass box with a built-in v-shaped trough (trough is covered with plastic cupboard lining to avoid mucus and scale loss). Plastic hoses are also connected to the surgery table for outtake and intake of water. The water from outtake hose is emptied into a 20 L bucket. (**Grant PUD provides**)
- Baking soda – sodium bicarbonate (buffering solution), same concentration as MS-222
- Stress Coat – original concentration and 25% solution 250 ml/L
- Syringes for measuring anesthetic (60 cc)
- Small and large dip nets – some modified as sanctuary nets
- Nitrile gloves
- Plastic weigh boats
- Trays for holding solutions used to clean surgical tools
- Needle drivers

- Forceps
- Disposable scalpels (Beaver micro-unitome knife, 15 degrees 5 mm)
- Germiphene (bactericidal, fungicidal, and virucidal; diluted 16 ml of Germiphene to 1 L of water)
- Chlorhexidine solution (Nolvasan; 30 ml/L) prepared with distilled water
- Sutures, size 5-0 (VICRYL coated with tapered RB-1 needle – antibacterial type)
- Distilled water as a rinse solution for disinfectant (Chlorhexidine)
- 2 L bottles to make up stock solutions for Stress Coat, Germiphene, and MS-222
- Spray bottles for Stress Coat and disinfectant
- Shop towels
- Sharps container and Ziploc Bags
- Autoclave (M11 Ultraclave, Midmark) (**LGL provides**)
- Data sheets, writing tools, computer (**LGL provides**)

B9 Culling Guidelines for Salmonid Smolts

A. Criteria for culling fish prior to a tagging session (fish sorters and taggers)

- Anesthetize all fish with MS-222 prior to handling.
- Keep fish underwater throughout the examination process, handling fish as quickly and gently as possible.
- Examine fish individually for external marks before tagging, including:
 - Scale loss: Normal (<4%); Partial (4-19%); De-scaled (>19%);



Scale loss is calculated per side. For example, 10% scale loss on each side would not be a reject, whereas 20% on one side would. Divide fish into ~ 25% portions as a reference when estimating scale loss. Example of normal (< 4%) scale loss (top) and partial scale loss (4-19%).

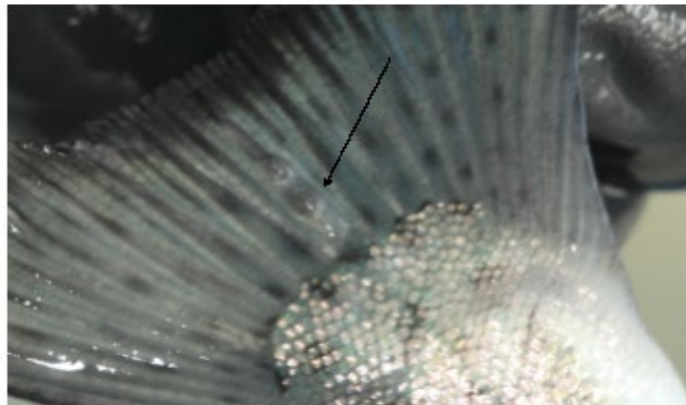
- Bird wounds, head and body injuries, net marks, bleeding, etc.



- Fungus, parasites, leeches, and overall fish health.



- Missing or extremely damaged fins (tattered fins are acceptable);
- Gas Bubble Trauma (GBT) symptoms (i.e., bubbles on fins)



- Severe abnormalities (stunted, duplicated fins, etc.);



- Signs of Bacterial Kidney Disease (BKD) include hemorrhaging near the eyes and bloating;



- Elastomer or visual implant (VI) marks, or other brands;



- PIT tag entry scars (between pectoral and ventral fins); and
- Cull all fish that have been previously anesthetized, handled, and tagged (except elastomer tagged fish).

iv. Size criteria (exclude):

- Salmonid with a FL that is less than 95 mm.
- Salmonid with a FL that is greater than 210 mm.
- Salmonid that weight less than 10.5g:

i. All tagged study fish will not exceed a 3% tag burden threshold,

v. Any fish that have been dropped or mishandled in any way

B. Criteria for culling fish during a tagging session (criteria for taggers only)

- Do not tag any fish you feel does not exhibit the same behavior as the group of fish being tagged during each tagging session. For example, if you have a fish that loses equilibrium in substantially less time (compared to the other fish you have been anesthetizing), there is probably a reason. The fish might have been handled recently or is unhealthy.
- Do not tag a fish that exhibits excess bleeding during tagging.
- Do not tag a fish that has excess body fluid once incised (sign of BKD).

C. Criteria for culling fish post-tagging (criteria for taggers only)

- i. If a tagged fish does not exhibit normal swimming behavior or has not gained equilibrium within 30 minutes of being tagged, it should be rejected. If the fish is rejected then the tag is pulled and put into another fish. The rejected fish should be re-sutured and released into the river.

D. Overall

- i. Keep an accurate tally of all rejected fish, and the reasoning used to reject it. Keep an accurate tally of all surplus fish.

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